AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 4 · WORKBOOK

Python and Table Readiness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 31

Recall and choose your starting point

> Record your starting explanation for Python and Table Readiness before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Classification or Regression?

Without opening the old lesson, explain one key idea from Classification or Regression?. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

The ML Lifecycle

Without opening the old lesson, explain one key idea from The ML Lifecycle. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 31

Practise with fading help

> Complete the supported Python and Table Readiness trace and record the first step you can explain without help.

Use W2: Classify records for Rows columns and **schema**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 31

Individual reasoning check

> Check the proposed Python and Table Readiness explanation against the supplied facts

before discussing it.

Use the worked case for Rows columns and **schema**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 31

Independent transfer case

Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Supplied values are3,7,14. Compute their mean. What should a function report for an empty list? How should it handle a missing fourth record?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 31

Your learning target

> Connect the Python and Table Readiness target to a trace and a justified conclusion.

Trace mean for [4,8,12] using sum(values)/len(values). What should the program do for []? Explain why returning 0 silently is misleading.

WORKBOOK · 6 OF 31

Solve the given case

> Solve the supplied Python and Table Readiness case and show the deciding evidence.

Python lengths=[4,6,8]; total=sum(lengths); mean=total/len(lengths).

Trace and handle an empty list.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 31

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Trace Python: lengths = [4, 7, 9]; selected = lengths[1]; total = selected + 3. Give selected and total.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 4 Test and reflect

WORKBOOK · 8 OF 31

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	

Test record	My evidence
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 31

Project checkpoint

> Justify what Python and Table Readiness adds to Responsible Plant Classifier and keep the evidence.

Create a tiny five-row plant table with explicit column names, **type**s, **unit**s, and a reproducible summary calculation.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 31

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 31

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Trace mean for [4,8,12] using sum(values)/len(values). What should the program do for []? Explain why returning 0 silently is misleading.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 31

W1 State a schema

> State a schema. Apply the supplied Python and Table Readiness facts and keep your reasoning.

List the required fields, **type**s and duration boundaries for the worked table.

My reasoning and evidence

W2 Classify records

> Classify records. Apply the supplied Python and Table Readiness facts and keep your reasoning.

Classify minutes values 0, 60, -1 and 61 under this **schema**.

My reasoning and evidence

W3 Repair a unit problem

> Repair a unit problem. Apply the supplied Python and Table Readiness facts and keep your reasoning.

A source says 180 seconds but the row says minutes: 180. Give a documented correction.

My reasoning and evidence

W4 Design a schema independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Design a schema independently. Apply the supplied Python and Table Readiness facts and keep your reasoning.

For a fictional shelf audit, define label and book_count fields. Require a nonempty text label and an integer count from 0 to 40. Give two valid and two invalid records with reasons.

My reasoning and evidence

W5 Separate validity and truth

> Separate validity and truth. Apply the supplied Python and Table Readiness facts and keep your reasoning.

A row says Reading, 12 and passes. Does that establish the session really lasted **12 minutes**?

My reasoning and evidence

File input and output — W1 Trace types

> Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

Before and after int("8"), identify the **type**s and values.

My reasoning and evidence

WORKBOOK · 18 OF 31

File input and output — W2 Predict the total

> Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

A sandbox **file** has lines 5, 0 and 2. What total does the worked reader print?

My reasoning and evidence

WORKBOOK · 19 OF 31

File input and output — W3 Repair a path issue

> Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

The **file** is in the class-data folder but the script looks in its working folder. Give a safe repair.

My reasoning and evidence

WORKBOOK · 20 OF 31

File input and output — W4 Read independently

> Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

In a fresh sandbox **file** counters.txt put 4, 6 and 0 on separate lines. Adapt the reader, print the number of values and total. State the expected results.

My reasoning and evidence

WORKBOOK · 21 OF 31

File input and output — W5 Handle malformed data

> Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

The **file** contains twelve on one line. Why should the program not silently turn it into zero?

My reasoning and evidence

Exceptions and recovery — W1 Trace success

Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

What does parse_minutes("0") return, and why is zero allowed?

My reasoning and evidence

Exceptions and recovery — W2 Trace two failures

> Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

Compare parse_minutes("abc") and parse_minutes("70").

My reasoning and evidence

Exceptions and recovery — W3 Repair a handler

> Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

An except block returns 0 for any failure. State two problems and a repair.

My reasoning and evidence

Exceptions and recovery — W4 Build independently

> Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

Adapt the function for count strings representing integers 0 through 10. Test "4", "0", "11", "four" and "-1" and record **status**/value.

My reasoning and evidence

Exceptions and recovery — W5 Choose recovery

> Use the supplied Python and Table Readiness task to record your reasoning, evidence and next step.

A batch has 8 valid and 2 invalid lines. What should its report include?

My reasoning and evidence

WORKBOOK · 27 OF 31

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 28 OF 31

Pictorial case practice: Python and Table Readiness

> Inspect a small table before asking Python to compute an average. Explain the labelled step before checking the changed condition.

Inspect a small table before asking Python to compute an average.

Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/4

Python and Table Readiness

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Check records

Four rows contain three measured values and one missing value.

2. Calculate mean

Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm.

3. Report coverage

Report 3/4 measured, retaining the missing record separately.

CHANGE ONE THING

S changes from 0 to None.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 29 OF 31

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 30 OF 31

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Inspect a small table before asking Python to compute an average.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Report 3/4 measured, retaining the missing record separately.

1. Check records

2. Report coverage

3. Calculate mean

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report coverage Report 3/4 measured, retaining the missing record separately. Read the picture in order. Report 3/4 measured, retaining the missing record separately.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: S changes from 0 to None. Which conclusion is supported by the

supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. Mean becomes (2+6)/2=4 cm with 2/4 measured.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Mean becomes (2+6)/2=4 cm with 2/4 measured. Removing a measured zero changes both the denominator and the coverage.

Repair a missing building block

Review Classification or Regression? (ai-4/3). Goal: Identify a task from the meaning of its target. Short practice: Did regression MAE improve when the boundary changed?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review The ML Lifecycle (ai-4/1). Goal: Distinguish **fitting** a rule from applying a fixed rule. Short practice: Should the team silently retune on the changed test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

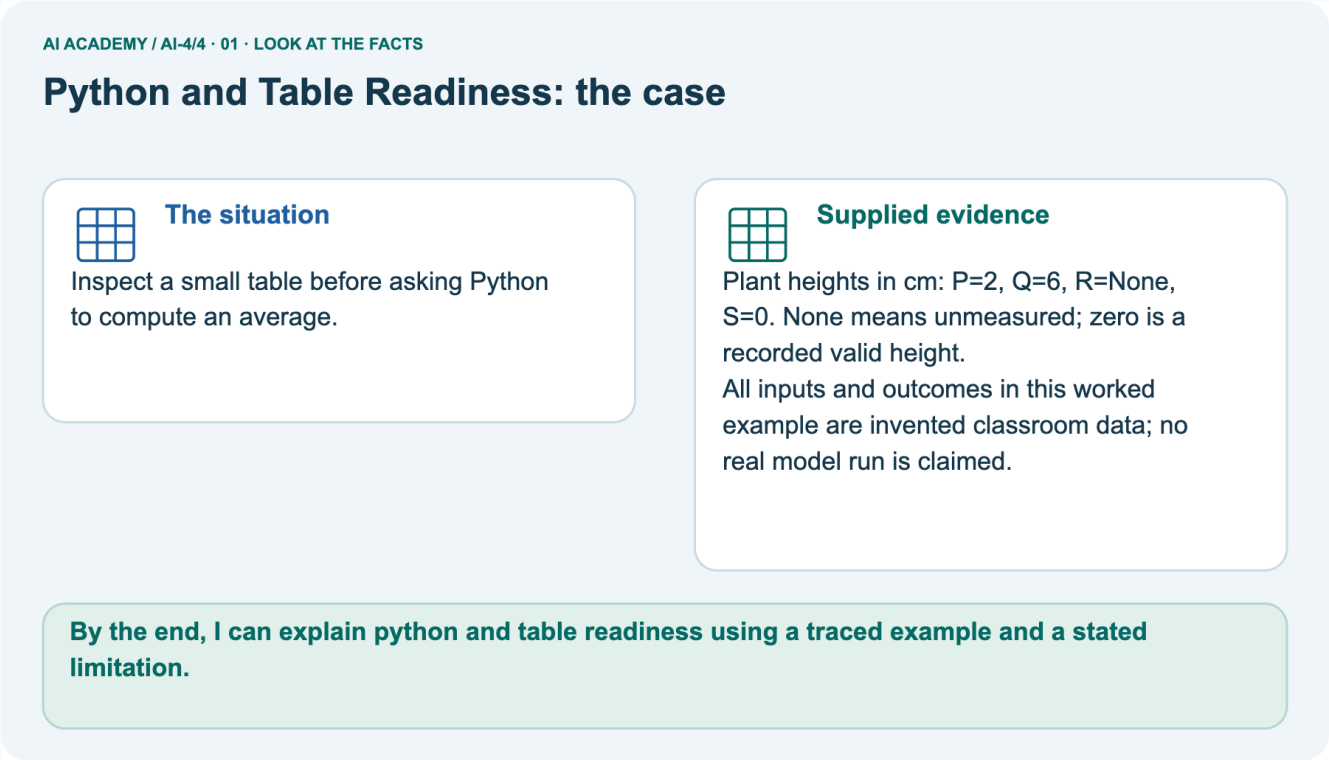
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Python and Table Readiness picture from its labels, then record a relevant limit.

01. Python and Table Readiness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

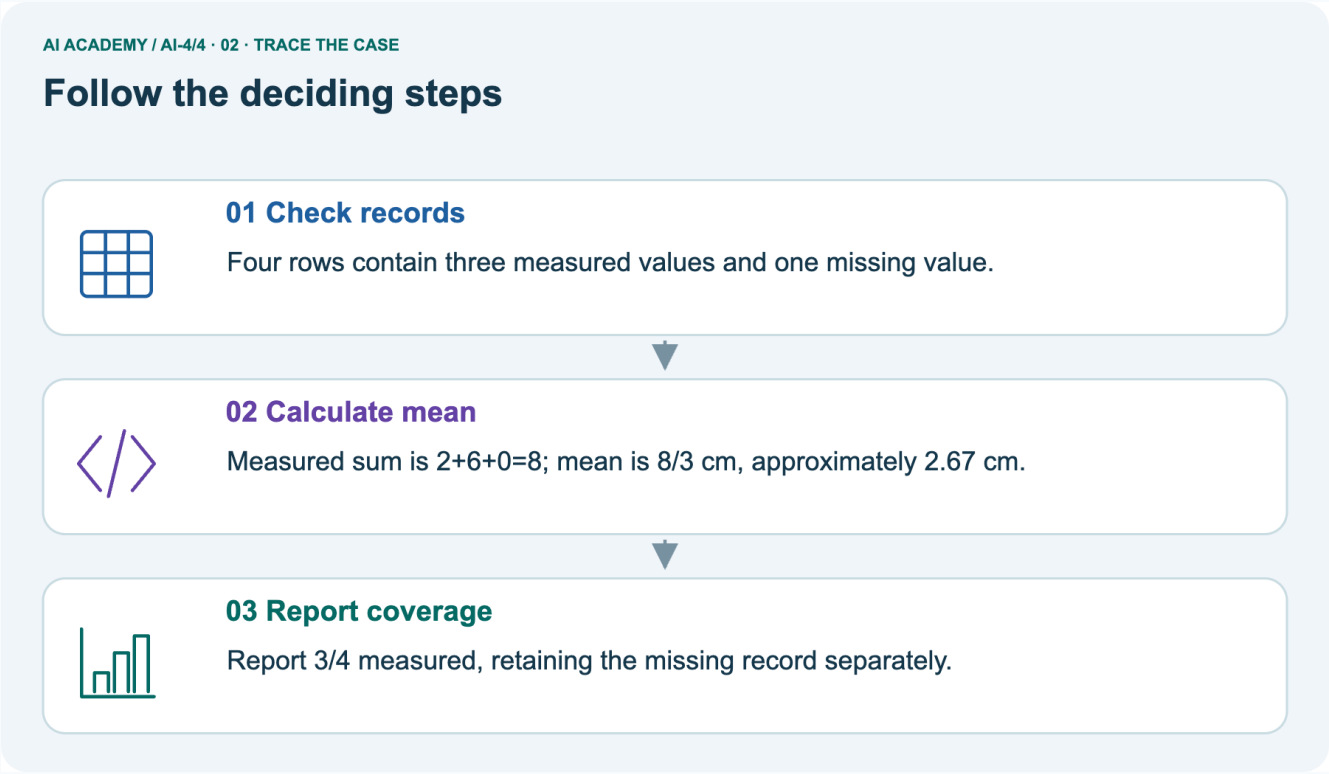
Inspect a small table before asking Python to compute an average. Supplied facts: Plant heights in cm: P=2, Q=6, R=None, S=0. None means unmeasured; zero is a recorded valid height.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



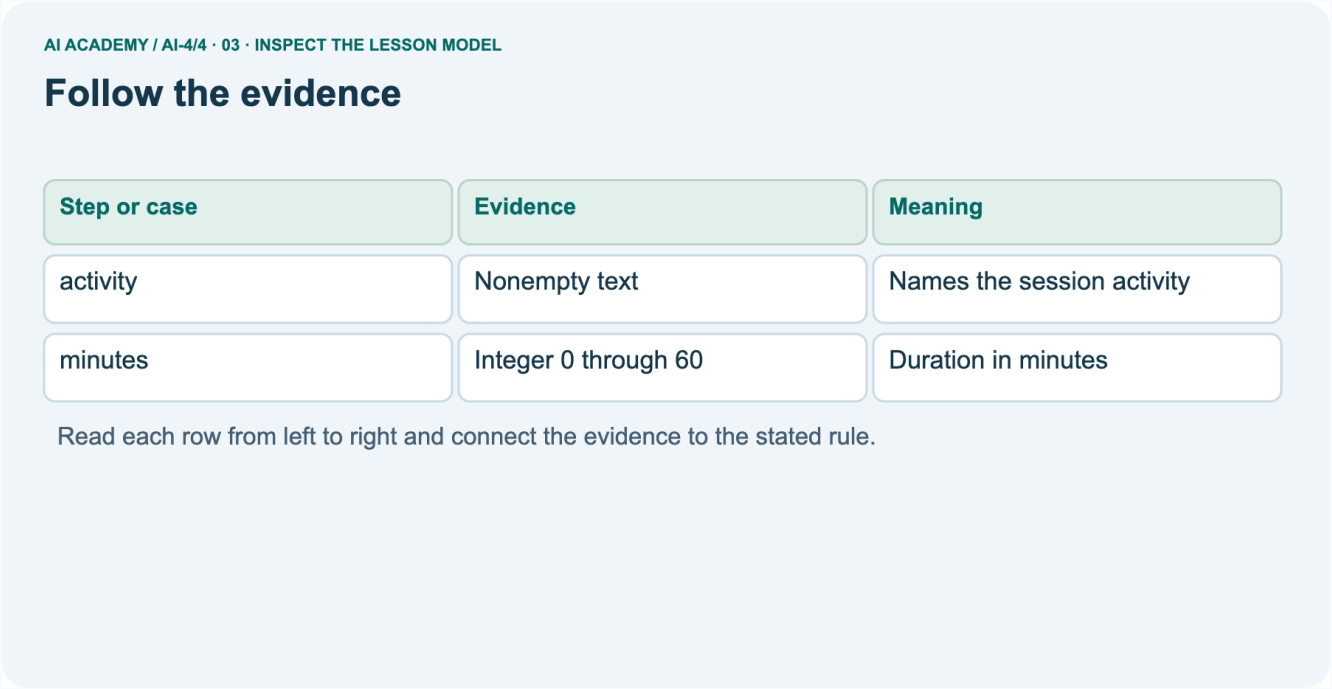
Follow the labelled picture with the matching explanation. Open the vector diagram

Check records: Four rows contain three measured values and one missing value. → Calculate mean: Measured sum is 2+6+0=8; mean is 8/3 cm, approximately 2.67 cm. → Report coverage: Report 3/4 measured, retaining the missing record separately.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



AI ACADEMY / AI-4/4 · 03 · INSPECT THE LESSON MODEL

Follow the evidence

Step or case	Evidence	Meaning
activity	Nonempty text	Names the session activity
minutes	Integer 0 through 60	Duration in minutes

Read each row from left to right and connect the evidence to the stated rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. activity | Nonempty text | Names the session activity. minutes | Integer 0 through 60 | Duration in minutes

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / AI-4/4 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Report 3/4 measured, retaining the missing record separately.

Changed condition

S changes from 0 to None.

Worked changed-case result

Mean becomes (2+6)/2=4 cm with 2/4 measured.
Why: Removing a measured zero changes both the denominator and the coverage.

Follow the labelled picture with the matching explanation. Open the vector diagram

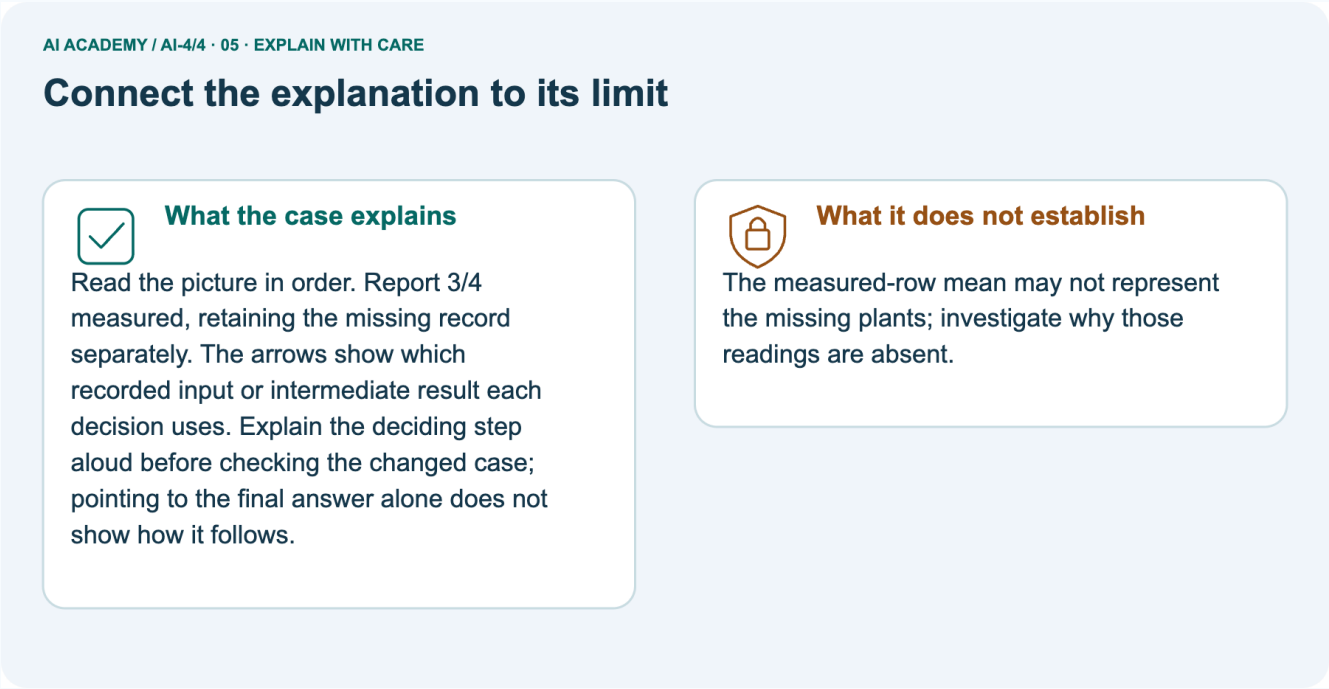
Original case: Report 3/4 measured, retaining the missing record separately. Changed condition: S changes from 0 to None. Expected result: Mean becomes (2+6)/2=4 cm with 2/4 measured.

Reason: Removing a measured zero changes both the denominator and the coverage.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Report 3/4 measured, retaining the missing record separately. The arrows show which recorded **input** or intermediate result each decision uses.

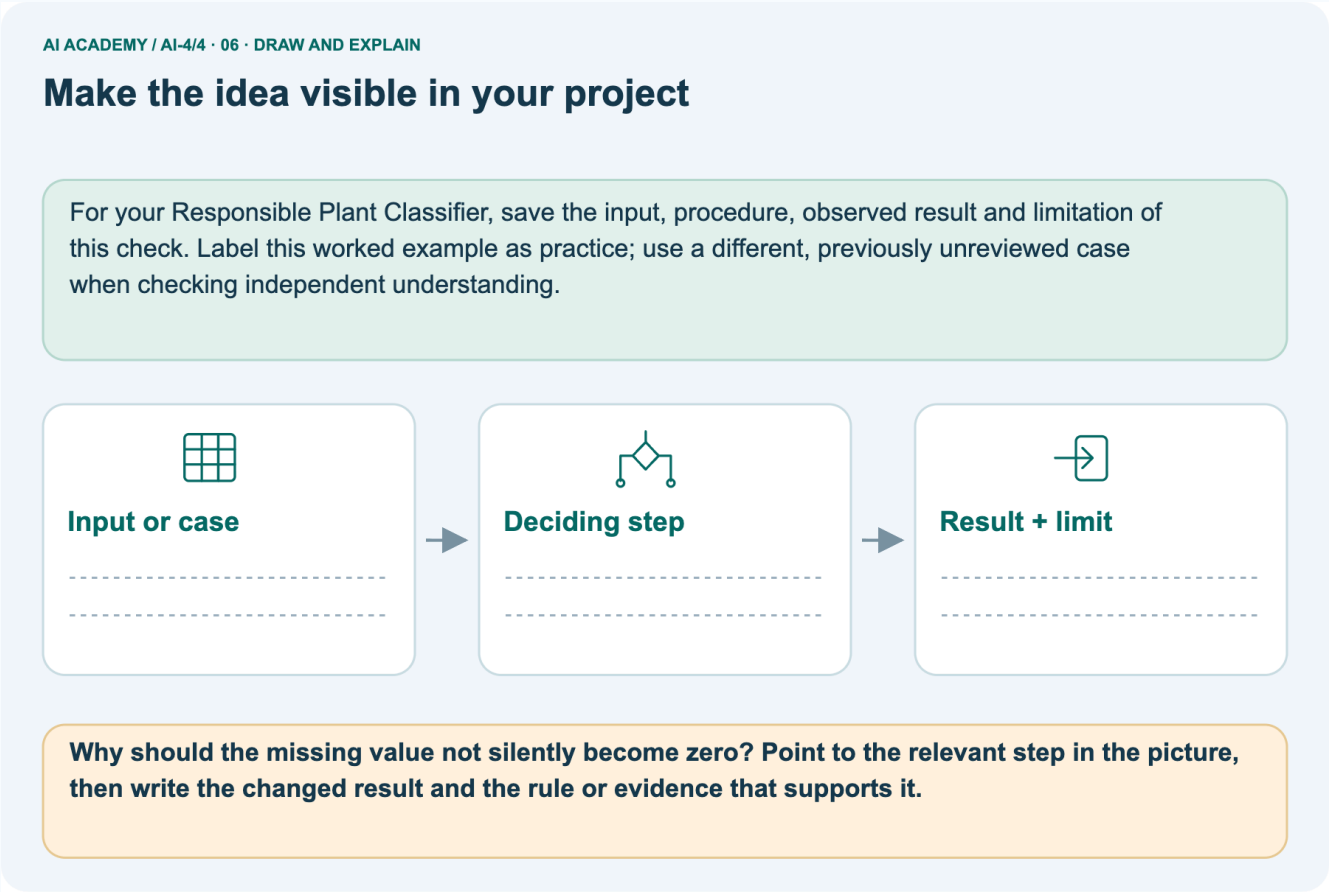
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: The measured-row mean may not represent the missing plants; investigate why those readings are absent.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the missing value not silently become zero? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.